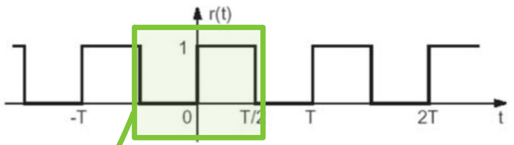
Exercise 3

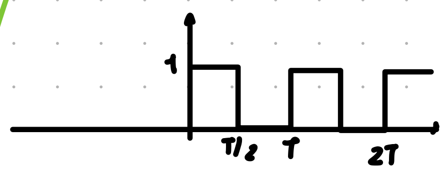
STEPS: • FIND DERIVATIVE
• SPLIT

Let r(t) be the rectangular, periodic wave function of period T in the figure:

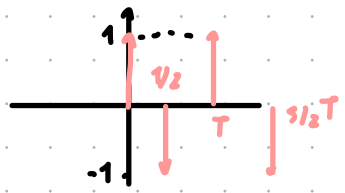


Calculate the coefficients of the Fourier series cn of its derivative (a) and explain why those of its integral (b) cannot be calculated. Finally, calculate the average value of r(t). (c)

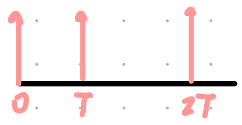
(a)



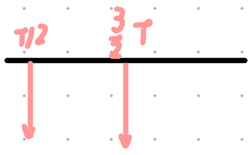
DERIVATE →



SPLIT



$c_n^{\oplus} = \frac{A}{T} = \frac{1}{T}$ ✓ CENTRED IN 0



$c_n^{\ominus} = \frac{A}{T} = -\frac{1}{T} e^{-j2\pi n \frac{T/2}{T}} = -\frac{1}{T} e^{-j\pi n}$

CENTRED IN T/2

PERIOD T

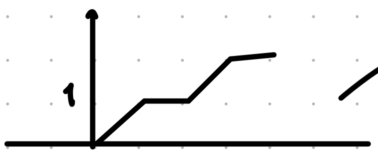
$c_n = c_n^{\oplus} + c_n^{\ominus} = \frac{1}{T} - \frac{1}{T} e^{-j\pi n} =$

$\boxed{e^{-i\pi} = -1}$

$= \frac{1}{T}\left(1-(-1)^n\right)$

$$c_n = \begin{cases} 2/T & \text{IF } n \text{ ODD} \\ 0 & \text{IF } n \text{ EVEN} \end{cases}$$

INTEGRAL ↓



IS NOT PERIODIC
CAN NOT CALCULATE c_n :^

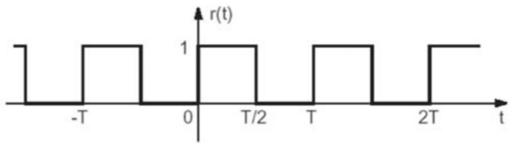
(3) AVERAGE VALUE OF r(t)

$$c_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} r(t)\, e^{-j\frac{2\pi n t}{T_0}} \longrightarrow c_0 = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} r(t)\cdot 1\, dt = \frac{1}{T_0}\, 0\Big|_{-T_0/2}^{0} + \frac{1}{T_0}\, t\Big|_{T_0/2}^{T_0} = \frac{1}{T_0}\left(T_0 - T_0/2\right) = \frac{1}{2}$$

$$\int r(t)\, dt = \begin{cases} 1\Big|_0^{-T_0/2} = 1-1=0 & \text{OTHERWISE} \\ x\Big|_{T_0/2}^{T} = T - T_0/2 = T_0/2 & nT/2 \leq x < nT \end{cases}$$

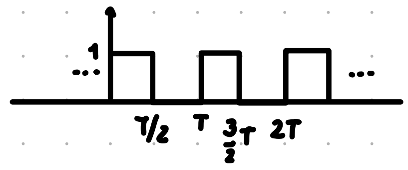
Exercise 3

Let r(t) be the rectangular, periodic wave function of period T in the figure:

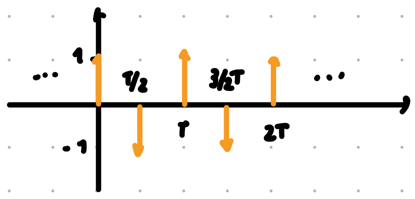


Calculate the coefficients of the Fourier series cn of its derivative and explain why those of its integral cannot be calculated. Finally, calculate the average value of r(t).

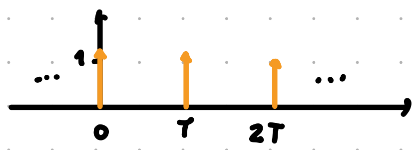
PERIOD T



DERIVATE THE SIGNAL

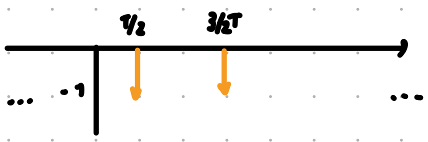


DIVIDE THE SIGNAL



$$C_n^{\uparrow} = \frac{A}{T} = \frac{1}{T} \underbrace{e^{\frac{-j2\pi n 0}{T}}}_{1}$$

CENTRED IN 0

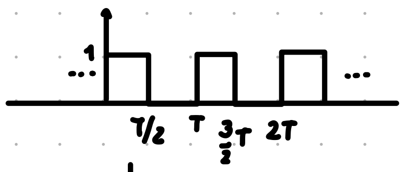


CENTRED IN 1/2 T

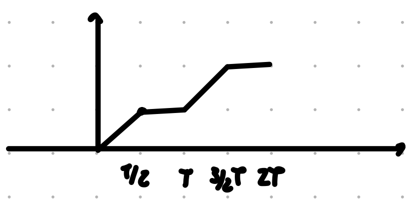
$$C_n^{\downarrow} = \frac{A}{T} = -\frac{1}{T} e^{\frac{-j2\pi n \frac{1}{2}T}{T}} = -\frac{1}{T} e^{-j\pi n}$$

$$e^{j\pi} = -1$$

$$C_n = C_n^{\uparrow} + C_n^{\downarrow} = \frac{1}{T} - \frac{1}{T} e^{-j\pi n} = \frac{1}{T} - \frac{1}{T}\left(e^{j\pi}\right)^{-n} = \frac{1}{T} - \frac{1}{T}(-1)^n = \frac{1}{T}\left(1-(-1)^n\right) \begin{cases} 0 & n \text{ EVEN} \\ \frac{2}{T} & n \text{ ODD} \end{cases}$$

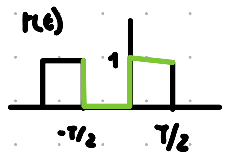


INTEGRATE



THE SIGNAL R(t) IS NOT PERIODIC
CAN NOT CALCULATE ITS COEFFICIENTS

$$C_n = \frac{1}{T}\int_{-T/2}^{T/2} r(t) e^{-\frac{j2\pi n t}{T}} dt \longrightarrow C_0 = \frac{1}{T}\int_{-T/2}^{T/2} r(t) e^{0} dt = \frac{1}{T}\int_{-T/2}^{0} r(t) dt + \frac{1}{T}\int_{0}^{T/2} r(t) dt = 0 + \frac{1}{2} = \boxed{\frac{1}{2}}$$



$$\longrightarrow r(t) = \begin{cases} 1 & \text{OTHERWISE} \\ 0 & \frac{n}{2}T < t < nT \end{cases}$$

$$* \quad \frac{1}{T}\int_{-T/2}^{0} r(t) dt = \frac{1}{T}\int_{-T/2}^{0} 0 \, dt = \frac{1}{T} t \Big|_{-T/2}^{0} = \frac{1}{T}\left[0 - (-T/2)\right] = \frac{1}{2}$$

SHOULD BE 0

$$* \quad \frac{1}{T}\int_{0}^{T/2} r(t) dt = \frac{1}{T}\int_{0}^{T/2} 1 \, dt = \frac{1}{T} t \Big|_{0}^{T/2} = \frac{1}{T}\left[T/2 - 0\right] = \frac{1}{2}$$